

Dynamics of the Stable Boundary Layer: A Geometric Singular Perturbation Framework for Fold Characterization and Regime Transitions

Part 1: Mathematical Foundations and Observational Resolution

David E. England, PhD

Research Engineer III

University of Alabama in Huntsville (UAH)

david.england@uah.edu

Manuscript prepared for SBL-GSPT collaborative research project stack.

Last compiled: July 31, 2026

Abstract

Observed critical Richardson thresholds vary widely across field campaigns (e.g., CASES-99 and SHEBA), yielding the long-standing Richardson threshold paradox. We show that this variability is not evidence of incompatible physics, but rather the projection of a multidimensional folded critical manifold onto scalar diagnostics. To formalize this mechanism, we construct a multiscale fast-slow model of the stable boundary layer using Geometric Singular Perturbation Theory (GSPT) on a desingularized coordinate chart. The framework yields a sequence of structural results: existence and smoothness of the critical manifold, fold characterization through fast-Jacobian degeneracy, and a constant-rank projection theorem linking invariant state-space geometry to campaign-dependent observational thresholds. Within this geometry, we define an analytical sensible heat-flux turning-capacity limiter, $H_{\max}$, and a pathway toward dynamic threshold parameterization. Part 1 establishes the mathematical foundations and observational resolution needed for closure development and data-assimilative validation in Parts 2–3.

1 Introduction

The nocturnal Stable Boundary Layer (SBL) remains one of the most persistent challenges in geophysical fluid dynamics due to the highly non-linear, non-equilibrium coupling between turbulent dissipation, surface energy budgets, and mesoscale forcing (Nieuwstadt, 1984; Mahrt, 2014; van de Wiel et al., 2017). A central unresolved problem in boundary-layer meteorology is the so-called “Richardson Number Universality Crisis”: the observation that the critical Bulk Richardson number (Ri_c), traditionally expected to be a quasi-universal constant near 0.25, varies wildly across different environmental regimes (Grachev et al., 2013; Monahan et al., 2015). While mid-latitude field campaigns such as CASES-99 report collapse thresholds near the classical limit ($Ri \approx 0.2$ – 0.4), polar campaigns like SHEBA record active turbulence persisting at stability ratios exceeding unity ($Ri > 1.2$) (Poulos et al., 2002; Grachev et al., 2005).

1.1 The Paradox of Stability Thresholds

For decades, the search for a “better” stability function has assumed that the variability in Ri_c stems from inconsistent physics or measurement noise. However, traditional parameterizations—such as Monin–Obukhov Similarity Theory (MOST) or Mellor–Yamada K -theory—rely on an instantaneous local equilibrium assumption, $e = e(S, N^2)$, that erases the system’s underlying temporal history and structural hysteresis (Mellor and Yamada, 1982). This diagnostic approach forces a smooth, single-valued response that cannot account for the abrupt, “brittle” transitions between weakly stable and very stable regimes.

1.2 Thesis: Richardson Thresholds as Projections, Not Invariants

In this paper, we propose that the Richardson threshold paradox is not evidence of incompatible physics, but rather a deterministic consequence of dimensionality reduction. We frame the SBL as a singularly perturbed dynamical system governed by Geometric Singular Perturbation Theory (GSPT) (Fenichel, 1979; Kuehn, 2015). Within this framework, our central thesis is that *Richardson thresholds are low-dimensional coordinate projections of a higher-dimensional folded equilibrium manifold*.

By exploiting the natural separation of timescales—where fast turbulence adjustment ($\tau_e \sim 10^1$ – 10^2 s) responds to slow surface thermal and geostrophic evolution ($\tau_{\text{slow}} \sim 10^3$ – 10^4 s)—we demonstrate that the “critical” threshold is not a static point, but a state-dependent fold locus ($\mathcal{C}_{\text{fold}}$) that deforms dynamically in response to surface energy balance (van de Wiel et al., 2012).

1.3 The “Fold Illusion” and Emergent Coupling

A key conceptual contribution of this work is the distinction between isolated boundary-driven decay and true fold catastrophes. We prove that while an isolated atmospheric subsystem exhibits only a monotonic boundary-crossing at the laminar floor ($e = 0$), a genuine S-shaped fold only emerges through the non-linear coupling of the atmosphere to the Surface Energy Balance (SEB). We term the historical conflation of these two mechanisms the “Fold Illusion,” and show that the geometric resolution of this illusion is necessary to model intermittent bursting and nocturnal relaxation oscillations accurately.

1.4 Objectives and Scope of Part 1

As the first in a three-part series, this manuscript establishes the mathematical foundations and observational resolution of the GSPT-SBL framework. We define a hierarchy of models—from 2D analytical baselines to 5D thermodynamic systems—and provide a sequence of structural theorems:

- **Theorem 1 (Existence and Smoothness):** Establishing the critical manifold ($\mathcal{S}_0$) as a smooth embedded submanifold of the state space.
- **Theorem 2 (Fold Characterization):** Defining the geometric boundary where normal hyperbolicity is lost, triggering turbulence collapse.
- **Theorem 3 (Projection Theorem):** Formalizing the mapping from invariant state-space geometry to campaign-specific scalar diagnostics (Ri_{fold}).

Part 1 provides the “geometric skeleton” required for the data-driven discovery and prognostic parameterizations developed in Parts 2 and 3.

- 1.5 The Crisis of Stability Thresholds in Boundary-Layer Meteorology
- 1.6 The Closure Trilemma in Atmospheric Modeling
- 1.7 Objectives and Scope of Part 1
- 2 Multi-Scale Governing Equations and Chart Regularization

3 Multi-Scale Governing Equations and Chart Regularization

This section defines the smooth fast-slow architecture for the stable boundary layer (SBL) model. We introduce the regularized state space, state the standing assumptions used in the geometric analysis, derive the desingularized fast chart, and define the observation map that connects invariant-manifold structure to atmospheric diagnostics.

3.1 System Definition and Standing Assumptions

We work with a multiscale autonomous system written in the desingularized time variable τ on an open chart domain $\Omega_0 \subset \mathbb{R}^5$:

$$\frac{d\mathbf{x}}{d\tau} = \mathbf{F}(\mathbf{x}; \boldsymbol{\mu}, \epsilon_1, \epsilon_2) \equiv \begin{pmatrix} F_{\text{fast}}(\mathbf{x}; \boldsymbol{\mu}) \\ \epsilon_1 F_{\text{slow}}(\mathbf{x}; \boldsymbol{\mu}) \\ \epsilon_1 \epsilon_2 F_{\text{superslow}}(\mathbf{x}; \boldsymbol{\mu}) \end{pmatrix}. \quad (1)$$

The state vector is

$$\mathbf{x} = \begin{pmatrix} \tilde{e} \\ q_\theta \\ S \\ T_s \\ T_g \end{pmatrix} \in \Omega_0 \subset \mathbb{R}_+ \times \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R}_+. \quad (2)$$

Its components are interpreted as follows: $\tilde{e} \in \mathbb{R}_+$ is the desingularized turbulent kinetic energy coordinate, $q_\theta \in \mathbb{R}$ is the kinematic turbulent heat flux $\overline{w'\theta'}$, $S \in \mathbb{R}_+$ is the bulk vertical wind shear $\|\partial \mathbf{u}/\partial z\|$, $T_s \in \mathbb{R}_+$ is the surface skin temperature, and $T_g \in \mathbb{R}_+$ is the deep-soil temperature.

Standing assumptions. Throughout this section we assume:

1. **(A1) Domain regularity.** The chart domain $\Omega_0 \subset \mathbb{R}^5$ is open and connected, and the vector field $\mathbf{F}$ is of class C^∞ on Ω_0 .
2. **(A2) Parameter domain.** The parameter block $\boldsymbol{\mu} \in \mathcal{P}$ belongs to an open bounded set of physically admissible coefficients containing the closure, radiative, and thermodynamic constants appearing below.
3. **(A3) Positive invariance.** The physically admissible set

$$\Omega_{\text{phys}} = \{\mathbf{x} \in \Omega_0 \mid \tilde{e} > 0, S > 0, T_s > 0, T_g > 0\} \quad (3)$$

is positively invariant under the flow.

4. **(A4) Local well-posedness.** For each initial state $\mathbf{x}(0) \in \Omega_0$, there exists a unique local trajectory $\mathbf{x}(\tau)$.
5. **(A5) Timescale separation.** The perturbation parameters satisfy

$$0 < \epsilon_1 \ll 1, \quad 0 < \epsilon_2 \ll 1, \quad (4)$$

with fast dynamics on $O(1)$ time, slow dynamics on $O(\epsilon_1^{-1})$ time, and super-slow dynamics on $O((\epsilon_1 \epsilon_2)^{-1})$ time.

Unless otherwise stated, we use Ω_0 to denote the full mathematical chart domain and Ω_{phys} to denote the positively invariant physical region used for geometric analysis. We further assume that Ω_0 extends across $\partial\Omega_{\text{phys}}$ to include a neighborhood of $\tilde{e} = 0$, so that $\mathcal{S}_0$ and $\mathcal{C}_{\text{fold}}$ can be analyzed as smooth embedded submanifolds near the laminar boundary.

3.2 Chart Regularization and Desingularization

In the original turbulent kinetic energy variable e , the fast equations contain non-smooth fractional powers through the shear-production term $c_m \ell e^{1/2} S^2$ and the Kolmogorov dissipation term $e^{3/2}/\ell$. As $e \rightarrow 0^+$, differentiability is lost at the laminar boundary.

To restore smoothness, introduce the regularized chart map

$$\phi_\delta : e \mapsto \tilde{e} = \sqrt{e + \delta}, \quad \delta > 0. \quad (5)$$

We formulate the geometric analysis on the regularized chart and use the limit $\delta \rightarrow 0^+$ as the asymptotic interpretation of the physical system. To desingularize the fast dynamics, use the positive time reparameterization

$$d\tau = \frac{dt}{\epsilon_1 \tilde{e}} \quad \Longleftrightarrow \quad \frac{dt}{d\tau} = \epsilon_1 \tilde{e}. \quad (6)$$

Proposition 1 (Regularization and diffeomorphic equivalence). *On the physical domain $e > 0$, the map ϕ_δ is a C^∞ diffeomorphism onto its image. Under the positive time change*

$$d\tau = \frac{dt}{\epsilon_1 \tilde{e}}, \quad (7)$$

the fast subsystem extends to a smooth vector field in the chart variable $\tilde{e}$, while preserving phase trajectories, equilibrium sets, and the signs of eigenvalues transverse to normally hyperbolic branches of the critical manifold.

Proof. For $\delta > 0$ and $e > 0$,

$$\phi_\delta(e) = \sqrt{e + \delta} \quad (8)$$

is smooth with derivative

$$\phi'_\delta(e) = \frac{1}{2\sqrt{e + \delta}} > 0, \quad (9)$$

so ϕ_δ is a smooth diffeomorphism onto its image. In the physical variable e , the fast TKE equation is

$$\epsilon_1 \frac{de}{dt} = c_m \ell e^{1/2} S^2 - \frac{g}{\theta_0} q_\theta - \frac{1}{\ell} e^{3/2}. \quad (10)$$

Since $\tilde{e}^2 = e + \delta$, differentiation gives

$$2\tilde{e} \frac{d\tilde{e}}{dt} = \frac{de}{dt}, \quad (11)$$

and hence

$$\frac{d\tilde{e}}{d\tau} = \frac{dt}{d\tau} \frac{d\tilde{e}}{dt} = \epsilon_1 \tilde{e} \cdot \frac{1}{2\tilde{e}} \frac{1}{\epsilon_1} \frac{de}{dt} = \frac{1}{2} \frac{de}{dt}. \quad (12)$$

Substituting the physical equation and using $e = \tilde{e}^2 - \delta$ yields

$$\frac{d\tilde{e}}{d\tau} = \frac{1}{2} \left(c_m \ell \sqrt{\tilde{e}^2 - \delta} S^2 - \frac{g}{\theta_0} q_\theta - \frac{1}{\ell} (\tilde{e}^2 - \delta)^{3/2} \right). \quad (13)$$

In the limiting chart $\delta \rightarrow 0^+$ this becomes

$$\frac{d\tilde{e}}{d\tau} = \frac{1}{2}c_m\ell S^2\tilde{e} - \frac{g}{2\theta_0}q_\theta - \frac{1}{2\ell}\tilde{e}^3, \quad (14)$$

which is polynomial in $\tilde{e}$. Because $d\tau/dt = \tilde{e}/\epsilon_1 > 0$ on Ω_{phys} , the reparameterization is orientation preserving. Standard results for positive time changes then imply preservation of orbits, equilibria, and the signs of eigenvalues transverse to normally hyperbolic branches. $\square$

3.3 Subsystem Vector Field Components

Under Assumption (A1) and Proposition 2.1, the desingularized vector field components are given explicitly as follows.

Fast subsystem. The fast subsystem $F_{\text{fast}}(\mathbf{x}) = (\tilde{F}(\mathbf{x}), \tilde{H}(\mathbf{x}))^T$ governs rapid turbulent adjustment:

$$F_{\text{fast}}(\mathbf{x}) = \begin{pmatrix} \frac{1}{2}c_m\ell S^2\tilde{e} - \frac{g}{2\theta_0}q_\theta - \frac{1}{2\ell}\tilde{e}^3 \\ -c_w\theta_z(T_s)\tilde{e}^3 - \frac{g}{\theta_0}c_\theta\ell q_\theta^2 - \frac{C_\theta}{\ell}\tilde{e}^2q_\theta \end{pmatrix}. \quad (15)$$

Here the local thermal stratification is represented by the smooth diagnostic map

$$\theta_z(T_s) = \frac{\theta_{\text{top}} - \Pi_s T_s}{\Delta z}. \quad (16)$$

Slow subsystem. The slow subsystem $F_{\text{slow}}(\mathbf{x}) = (F_S(\mathbf{x}), F_{T_s}(\mathbf{x}))^T$ governs wind shear and surface thermal evolution:

$$F_{\text{slow}}(\mathbf{x}) = \begin{pmatrix} \tilde{e} \left[\mathcal{F}_{\text{ls}} - \frac{\partial}{\partial z}(c_m\ell\tilde{e}S) \right] \\ \frac{\tilde{e}}{C_s} \left[R_{\text{net}}(T_s) + \rho c_p q_\theta + \frac{k_g}{d_g}(T_g - T_s) \right] \end{pmatrix}, \quad (17)$$

where the factors $\tilde{e}$ arise from the positive time reparameterization, so the slow tendencies freeze in fast time as $\tilde{e} \rightarrow 0^+$. The corresponding net radiation closure is

$$R_{\text{net}}(T_s) = R_{\text{sw}}^\downarrow + \epsilon_a \sigma T_a^4 - \epsilon_s \sigma T_s^4. \quad (18)$$

Super-slow subsystem. The super-slow subsystem governs deep-soil heat conduction:

$$F_{\text{superslow}}(\mathbf{x}) = \tilde{e} \frac{k_g}{d_g^2}(T_s - T_g). \quad (19)$$

As in the slow subsystem, the prefactor $\tilde{e}$ is induced by the positive time change; all geometric arguments in subsequent sections are restricted to Ω_{phys} .

3.4 Observation Mapping and Diagnostic Functionals

To connect the internal state $\mathbf{x} \in \Omega_0$ to observable diagnostics, define the smooth observation map

$$\mathbf{\Pi}_{\text{obs}} : \Omega_0 \rightarrow \mathcal{O} \subset \mathbb{R}^3, \quad (20)$$

where $\mathcal{O}$ denotes the space of smooth diagnostic quantities accessible from field observations or numerical model output. Its components are

$$\mathbf{\Pi}_{\text{obs}}(\mathbf{x}) = \begin{pmatrix} \pi_{Ri}(\mathbf{x}) \\ \pi_H(\mathbf{x}) \\ \pi_S(\mathbf{x}) \end{pmatrix} = \begin{pmatrix} \frac{g}{\theta_0} \frac{\theta_z(T_s)}{S^2} \\ \rho c_p q_\theta \\ S \end{pmatrix}. \quad (21)$$

These functionals represent the gradient Richardson diagnostic, the sensible heat flux, and the wind-shear coordinate, respectively.

Differential structure of π_{Ri} . The differential covector of the Richardson functional is

$$D\pi_{Ri}(\mathbf{x}) = \begin{pmatrix} 0 & 0 & -\frac{2g\theta_z(T_s)}{\theta_0 S^3} & \frac{g}{\theta_0 S^2} \frac{\partial \theta_z}{\partial T_s} & 0 \end{pmatrix}. \quad (22)$$

Hence $D\pi_{Ri}$ lies entirely in the slow cotangent subspace $\text{span}\{dS, dT_s\}$, so the Richardson diagnostic is insensitive, at first order, to perturbations in the fast turbulent directions $(\tilde{e}, q_\theta)$.

Proposition 2 (Observation smoothness). *Under Assumptions (A1) and (A3), the map $\mathbf{\Pi}_{\text{obs}}$ and its component functionals (π_{Ri}, π_H, π_S) are C^∞ on Ω_0 .*

Proof. Each component of $\mathbf{\Pi}_{\text{obs}}$ is a composition of smooth algebraic operations on Ω_0 . Because $S > 0$ on Ω_{phys} , the denominator in

$$\pi_{Ri}(\mathbf{x}) = \frac{g}{\theta_0} \frac{\theta_z(T_s)}{S^2} \quad (23)$$

never vanishes. The remaining components are linear in q_θ and S , and $\theta_z(T_s)$ is smooth by construction. Therefore $\mathbf{\Pi}_{\text{obs}} \in C^\infty(\Omega_0, \mathcal{O})$. $\square$

Remark 2.2 (Geometry versus observation). The dynamical system evolves on the five-dimensional state manifold Ω_0 , whereas atmospheric diagnostics are represented in observation space through the smooth map

$$\mathbf{\Pi}_{\text{obs}} : \Omega_0 \rightarrow \mathcal{O}. \quad (24)$$

Consequently, invariant geometric objects in state space need not appear as invariant objects in observation space.

Section 2 now provides a smooth desingularized dynamical system, an explicit regularized chart, and a diagnostic observation framework suitable for the invariant-manifold results developed in the subsequent sections.

4	Critical Manifold $\mathcal{S}_0$ and Fold Characterization
4.1	Theorem 1: Existence and Smoothness of the Critical Manifold
4.2	Theorem 2: Fold Characterization Theorem
4.3	The Fold Illusion versus Emergent Coupled Catastrophes
5	Projection Theorem and Resolution of the Richardson Paradox
5.1	Diagnostic Projection Mapping π_{Ri}
5.2	Theorem 3: Projection Theorem
5.3	Environmental Constraint Manifolds Σ_{site} and Site Profiles
5.4	Proposition 3.2: Observational Identifiability
5.5	Reconciling Field-Campaign Observations
6	Geometric Taxonomy of Regime Transitions and Nocturnal Life-cycles
6.1	Dimensional-Reduction Hierarchy
6.2	Canonical Singularities and Intermittency
6.3	Four-Phase Nocturnal Relaxation Oscillation
7	Framework for Next-Generation Model Closures
7.1	Analytical H_{max} Sensible Heat-Flux Limiter
7.2	Dynamic Threshold Parameterization $Ri_{\text{fold}}(T_s, T_g)$
7.3	Adaptive Boundary-Layer Scale Heights h_{eff}
8	Conclusions and Roadmap
8.1	Summary of Theoretical Contributions
8.2	Roadmap for Parts 2 and 3
A	Explicit Determinant Derivation of the Fast Jacobian J_f
B	Derivation of the H_{max} Heat-Flux Turning Capacity Formula
C	Proof of Transversality for Environmental Constraint Manifolds

References

- Neil Fenichel. Geometric singular perturbation theory for ordinary differential equations. *Journal of Differential Equations*, 31(1):53–98, 1979. doi: 10.1016/0022-0396(79)90152-9.
- Christopher K. R. T. Jones. Geometric singular perturbation theory. In Italy Montecatini Terme, editor, *Dynamical Systems: Lectures Given at the 2nd Session of the Centro Internazionale Matematico Estivo (C.I.M.E.)*, volume 1609 of *Lecture Notes in Mathematics*, pages 44–118. Springer, Berlin, Heidelberg, 1995. doi: 10.1007/BFb0094293.
- Martin Krupa and Peter Szmolyan. Extending geometric singular perturbation theory to nonhyperbolic points—Fold and canard points in two dimensions. *SIAM Journal on Mathematical Analysis*, 33(2):286–314, 2001. doi: 10.1137/S0036141099360919.
- Nils Berglund and Barbara Gentz. Geometric singular perturbation theory for stochastic differential equations. *Journal of Differential Equations*, 191(1):1–54, 2003. doi: 10.1016/S0022-0396(03)00020-2.
- Edward N. Lorenz. On the existence of a slow manifold. *Journal of the Atmospheric Sciences*, 43(15):1547–1558, 1986. doi: 10.1175/1520-0469(1986)043<1547:OTEOAS>2.0.CO;2.
- Peter De Maesschalck and Mathieu Desroches. Numerical continuation techniques for planar slow-fast systems. *SIAM Journal on Applied Dynamical Systems*, 12(3):1159–1180, 2013. doi: 10.1137/120877386.
- Christian Kuehn. *Multiple Time Scale Dynamics: An Introduction Specializing to Differential Equations*, volume 191 of *Applied Mathematical Sciences*. Springer International Publishing, Cham, Switzerland, 2015. ISBN 978-3-319-12315-8. doi: 10.1007/978-3-319-12315-8.
- Samuel Jelbart and Christian Kuehn. Discrete geometric singular perturbation theory. *Discrete and Continuous Dynamical Systems - Series A*, 43(1):57–120, 2023. doi: 10.3934/dcds.2022142.
- Kristian Uldall Kristiansen. A review of multiple-time-scale dynamics: Fundamental phenomena and mathematical methods. In Bernhelm Booß-Bavnbek, Jens Christensen, Katherine Richardson, and Oriol Codina, editors, *Multiplicity of Time Scales in Complex Systems: Challenges for Sciences and Communication II*, pages 309–363. Springer Nature Switzerland, Cham, 2024. ISBN 978-3-031-45105-8. doi: 10.1007/16618_2023_75.
- Richard T. McNider and Roger A. Pielke. Diurnal boundary-layer development over sloping terrain. *Journal of the Atmospheric Sciences*, 38(10):2198–2212, 1981. doi: 10.1175/1520-0469(1981)038<2198:DBLDOS>2.0.CO;2.
- George L. Mellor and Tetsuji Yamada. Development of a turbulence closure model for geophysical fluid problems. *Reviews of Geophysics*, 20(4):851–875, 1982. doi: 10.1029/RG020i004p00851.
- F. T. M. Nieuwstadt. The turbulent structure of the stable atmospheric boundary layer. *Journal of the Atmospheric Sciences*, 41(14):2202–2216, 1984. doi: 10.1175/1520-0469(1984)041<2202:TTSOTS>2.0.CO;2.
- David E. England and Richard T. McNider. Stability functions based upon shear functions. *Boundary-Layer Meteorology*, 74(1-2):113–130, 1995. doi: 10.1007/BF00715713.

- Richard T. McNider, David E. England, et al. Predictability of the stable atmospheric boundary layer. *Journal of the Atmospheric Sciences*, 52(10):1602–1614, 1995. doi: 10.1175/1520-0469(1995)052<1602:POTSAB>2.0.CO;2.
- X. Shi, Richard T. McNider, et al. On the behavior of the stable boundary layer and role of initial conditions. *Pure and Applied Geophysics*, 162(10):1811–1829, 2005. doi: 10.1007/S00024-005-2694-7.
- J. Cuxart, A. A. M. Holtslag, et al. Single-column model intercomparison for a stably stratified atmospheric boundary layer. *Boundary-Layer Meteorology*, 118(2):273–303, 2006. doi: 10.1007/s10546-005-3780-1.
- Sergej Zilitinkevich, Tov Elperin, Nathan Kleeorin, and Igor Rogachevskii. Energy- and flux-budget (EFB) turbulence closure model for stably stratified flows. Part I: Steady-state, homogeneous regimes. *Boundary-Layer Meteorology*, 125(2):167–192, 2007. doi: 10.1007/s10546-007-9189-2.
- Sukanta Basu, A. A. M. Holtslag, B. J. H. van de Wiel, et al. Evaluating the performance of large-eddy simulations in stably stratified boundary layers. *Boundary-Layer Meteorology*, 128(1): 117–135, 2008. doi: 10.1007/s10546-008-9271-x.
- B. J. H. van de Wiel, A. F. Moene, et al. The minimum wind speed for sustainable turbulence in the nocturnal boundary layer. *Journal of the Atmospheric Sciences*, 69(11):3116–3127, 2012. doi: 10.1175/JAS-D-12-0107.1.
- Andrey A. Grachev, Edgar L. Andreas, et al. The critical Richardson number and turbulence in stably stratified boundary layers. *Boundary-Layer Meteorology*, 147(1):23–42, 2013. doi: 10.1007/s10546-012-9772-8.
- Larry Mahrt. Stably stratified atmospheric boundary layers. *Annual Review of Fluid Mechanics*, 46(1):23–45, 2014. doi: 10.1146/annurev-fluid-010313-141354.
- Adam H. Monahan, T. Rees, Yanping He, and Norman A. McFarlane. Multiple regimes of wind, stratification, and turbulence in the stable boundary layer. *Journal of the Atmospheric Sciences*, 72(8):3178–3198, 2015. doi: 10.1175/JAS-D-14-0311.1.
- B. J. H. van de Wiel, É. Vignon, Peter Baas, et al. Regime transitions in near-surface temperature inversions: A conceptual model. *Journal of the Atmospheric Sciences*, 74(4):1057–1073, 2017. doi: 10.1175/JAS-D-16-0180.1.
- J. van Hooft, P. Baas, M. van Tiggelen, C. Ansorge, and B. J. H. van de Wiel. An idealized description for the diurnal cycle of the dry atmospheric boundary layer. *Journal of the Atmospheric Sciences*, 76(12):3717–3736, 2019. doi: 10.1175/JAS-D-19-0023.1.
- Elizabeth Ramsey and Adam H. Monahan. Empirical low-dimensional dynamics of atmospheric stable boundary layer temperature inversions. *Journal of the Atmospheric Sciences*, 79(7):1965–1984, 2022. doi: 10.1175/JAS-D-21-0205.1.
- W. Shaw, L. Berg, M. Debnath, et al. Scientific challenges to characterizing the wind resource in the marine atmospheric boundary layer. *Wind Energy Science*, 7(6):2307–2334, 2022. doi: 10.5194/wes-2021-156.
- Jun-Yi He and Qiu-Sheng Li. *Winds in the Atmospheric Boundary Layer*. Springer Nature Singapore, Singapore, 2025. ISBN 978-981-95-1516-5. doi: 10.1007/978-981-95-1517-2.

- Otávio Acevedo and Larry Mahrt. Systematic vertical variation of mesoscale fluxes in the nocturnal boundary layer. *Boundary-Layer Meteorology*, 135(1):19–30, 2010. doi: 10.1007/s10546-010-9465-4.
- Gregory S. Poulos, William Blumen, David C. Fritts, et al. CASES-99: A comprehensive investigation of the stable atmospheric boundary layer. *Bulletin of the American Meteorological Society*, 83(4): 555–581, 2002. doi: 10.1175/1520-0477(2002)083<0555:CACIOT>2.3.CO;2.
- Andrey A. Grachev, Christopher W. Fairall, et al. Stable boundary-layer scaling regimes: The SHEBA data. *Boundary-Layer Meteorology*, 116(2):201–235, 2005. doi: 10.1007/s10546-004-2729-0.
- É. Vignon, F. Hourdin, C. Genthon, et al. Antarctic boundary layer parametrization in a general circulation model: 1-D simulations facing summer observations at Dome C. *Journal of Geophysical Research: Atmospheres*, 122(13):6818–6843, 2017. doi: 10.1002/2017JD026802.
- Lian Su, Chunsong Lu, Jinlong Yuan, Xiaofei Wang, Qing He, and Haiyun Xia. Measurement report: The promotion of the low-level jet and thermal effects on the development of the deep convective boundary layer at the southern edge of the Taklimakan Desert. *Atmospheric Chemistry and Physics*, 24(18):10947–10963, 2024. doi: 10.5194/acp-24-10947-2024.
- Lu Zhang, Hongsheng Zhang, Xuhui Cai, Yu Song, and Xiaoye Zhang. Characteristics of turbulence intermittency, fine structure, and flux correction in the Taklimakan desert. *Journal of the Atmospheric Sciences*, 81(2):459–475, 2024. doi: 10.1175/JAS-D-23-0107.1.
- Ernst Hairer and Gerhard Wanner. *Solving Ordinary Differential Equations II: Stiff and Differential-Algebraic Problems*. Springer Series in Computational Mathematics. Springer-Verlag, Berlin, 2nd edition, 2010. doi: 10.1007/978-3-642-05221-7.
- Bernd Krauskopf, Hinke Osinga, and Jorge Galán-Vioque. *Numerical Continuation Methods for Dynamical Systems: Path-Following and Boundary Value Problems*. Springer, 2007. doi: 10.1007/978-1-4020-6356-5.
- Christopher Rackauckas and Qing Nie. DifferentialEquations.jl – A performant and feature-rich ecosystem for solving differential equations in Julia. *Journal of Open Research Software*, 5(1):15, 2017. doi: 10.5334/jors.151.
- Gerd Steinebach. Construction of Rosenbrock–Wanner method Rodas5P and numerical benchmarks within the Julia Differential Equations package. *BIT Numerical Mathematics*, 63(2):27, 2023. doi: 10.1007/s10543-023-00967-x.